

A Two-Stage Deep Learning Framework for Underwater Fish Segmentation and Multi-Object Tracking via Knowledge Transfer

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Abstract—Accurate underwater fish segmentation and multi-object tracking are essential for marine ecosystem monitoring, fisheries management, and behavioral analysis. However, underwater environments present significant challenges due to poor illumination, color attenuation, scattering effects, dynamic backgrounds, and frequent object occlusions. To address these challenges, this paper proposes a two-stage deep learning framework that establishes knowledge transfer between fish segmentation and multi-object tracking. In the first stage, a segmentation network is developed using a ConvNeXt-Small encoder integrated with a Variational Autoencoder (VAE), learned positional encoding, multi-head self-attention, attention-gated skip connections, and a U-Net decoder to generate accurate fish segmentation masks from the Fish4Knowledge dataset. The attention gates selectively emphasize informative encoder features while suppressing irrelevant background information, leading to improved segmentation performance. In the second stage, the pretrained ConvNeXt encoder is transferred to the tracking framework as an appearance feature extractor. Fish are detected using Faster R-CNN on GMOT40 image sequences, and appearance embeddings are combined with Intersection over Union (IoU)-based spatial similarity for robust data association using the Hungarian algorithm. By reusing the learned feature representations from the segmentation stage, the proposed framework reduces redundant feature learning while improving tracking consistency. Experimental results demonstrate the effectiveness of the proposed framework for accurate underwater fish segmentation and reliable multi-object tracking under challenging underwater conditions.

Index Terms—Underwater Fish Segmentation, Multi-Object Tracking, ConvNeXt, Variational Autoencoder, Self-Attention, Knowledge Transfer

I. INTRODUCTION

Underwater fish monitoring plays a crucial role in marine ecosystem conservation, fisheries management, biodiversity assessment, and autonomous underwater exploration. Automated computer vision techniques have become increasingly important for analyzing underwater imagery, reducing manual effort while enabling continuous observation of marine life. Among these tasks, fish segmentation and multi-object tracking are fundamental for accurate fish localization, behavioral analysis, and population monitoring [1], [11], [12].

Despite recent advances in deep learning, underwater image analysis remains challenging due to low illumination, color

distortion, scattering effects, suspended particles, and complex backgrounds. In addition, fish exhibit significant variations in size, shape, pose, and appearance, often accompanied by occlusions and motion blur. These factors make accurate segmentation difficult and also affect the reliability of object tracking across consecutive frames.

Encoder-decoder architectures have achieved promising results for underwater image segmentation [1]. However, conventional convolutional networks mainly capture local spatial information and often struggle to model uncertainty and long-range contextual dependencies in complex underwater scenes. Similarly, most multi-object tracking methods learn appearance representations independently from the segmentation stage, resulting in redundant feature learning and limited utilization of segmentation knowledge [3]–[5].

To address these challenges, this paper proposes a two-stage deep learning framework for underwater fish segmentation and multi-object tracking through knowledge transfer. In the first stage, a segmentation network based on a ConvNeXt-Small encoder [3], Variational Autoencoder (VAE) [4], learned positional encoding, multi-head self-attention [5], and a U-Net decoder [1] is trained on the Fish4Knowledge dataset to generate accurate fish segmentation masks. In the second stage, the pretrained ConvNeXt encoder is reused as an appearance feature extractor for multi-object tracking on the GMOT40 dataset. Faster R-CNN [6] is employed for fish detection, while appearance similarity and Intersection over Union (IoU) are jointly utilized with the Hungarian algorithm [7] for robust data association across image sequences.

The main contributions of this work are summarized as follows:

- We propose a two-stage deep learning framework that connects underwater fish segmentation and multi-object tracking through knowledge transfer.
- We develop a segmentation network by integrating ConvNeXt-Small, a Variational Autoencoder, learned positional encoding, multi-head self-attention, and a U-Net decoder for robust underwater fish segmentation.
- We transfer the pretrained segmentation encoder to the tracking stage as an appearance feature extractor, reduc-

ing redundant feature learning and improving identity association.

- We develop a hybrid tracking framework that combines Faster R-CNN detection, appearance similarity, IoU-based matching, and the Hungarian algorithm for reliable multi-object fish tracking.

II. RELATED WORK

A. Underwater Fish Segmentation

Deep learning-based encoder-decoder architectures have significantly improved underwater image segmentation by learning discriminative pixel-level representations. Conventional U-Net models effectively combine encoder and decoder features through skip connections, enabling accurate localization of foreground objects [1]. However, underwater environments introduce challenges such as low visibility, color attenuation, backscattering, dynamic backgrounds, and object deformation, which often reduce segmentation performance. To better preserve spatial and contextual information, several studies have explored advanced encoder backbones and feature refinement strategies. Kapoor *et al.* proposed an encoder-decoder framework with a ResNet-50 backbone and GraphSAGE to restore latent feature relationships for underwater moving object detection, demonstrating improved segmentation performance under challenging underwater conditions [18].

B. ConvNeXt and Attention Mechanisms

Recent advances in convolutional architectures have shown that modernized convolutional networks can achieve performance comparable to transformer-based models. ConvNeXt introduces improved architectural designs that enhance feature representation while maintaining the efficiency and simplicity of convolutional networks [3].

Attention mechanisms have also become an effective solution for improving segmentation accuracy. Attention U-Net replaces conventional skip connections with attention gates that selectively emphasize informative encoder features while suppressing irrelevant background responses before feature fusion. This mechanism enables the decoder to focus on salient object regions and improves segmentation performance without introducing significant computational overhead [2].

C. Underwater Multi-Object Tracking

Multi-object fish tracking remains challenging because of similar fish appearances, frequent occlusions, rapid motion, and degraded underwater image quality. Existing tracking frameworks commonly combine object detection, appearance modeling, and data association to maintain consistent object identities. Mukherjee *et al.* proposed a Siamese-network-based tracking framework integrated with underwater image enhancement to improve appearance similarity estimation [17]. Their method combines appearance similarity, motion prediction, and IoU-based matching using the Hungarian algorithm to generate robust fish trajectories under challenging underwater environments [17].

D. Research Gap

Although existing segmentation and tracking methods have achieved encouraging performance, they are generally developed as independent tasks. Most segmentation models focus only on improving pixel-level accuracy, while tracking frameworks usually learn appearance features separately, resulting in redundant feature extraction. Moreover, conventional U-Net skip connections directly propagate encoder features to the decoder without selectively suppressing irrelevant background information. Motivated by these limitations, this work proposes a two-stage framework that integrates a ConvNeXt-Small encoder, Variational Autoencoder, learned positional encoding, attention-gated skip connections, and multi-head self-attention for robust underwater fish segmentation. Furthermore, the pretrained encoder is transferred to the tracking stage as an appearance feature extractor, establishing an effective knowledge transfer strategy between segmentation and multi-object tracking.

III. PROPOSED TWO-STAGE UNDERWATER FISH ANALYSIS FRAMEWORK

A. Overall Framework

The proposed framework consists of two sequential stages designed to perform accurate underwater fish segmentation and reliable multi-object tracking through knowledge transfer. In the first stage, a deep segmentation network is trained using underwater images from the Fish4Knowledge dataset to learn discriminative fish representations while generating precise binary segmentation masks. The network integrates a ConvNeXt-Small encoder, a Variational Autoencoder (VAE), learned positional encoding, multi-head self-attention, attention-gated skip connections, and a U-Net decoder to effectively capture both local and global contextual information under challenging underwater conditions.

After the segmentation network is trained, the learned ConvNeXt encoder is transferred to the tracking stage as an appearance feature extractor. Instead of training a separate feature representation network, the transferred encoder provides discriminative appearance embeddings for detected fish. In the second stage, Faster R-CNN performs fish detection on GMOT40 frame sequences, while appearance similarity and Intersection over Union (IoU) are jointly utilized for object association using the Hungarian assignment algorithm. This two-stage design establishes an effective knowledge transfer strategy that reduces redundant feature learning while improving tracking robustness and identity preservation across consecutive frames.

The overall workflow of the proposed framework is illustrated in Fig.1.

B. Stage I: Underwater Fish Segmentation

The first stage of the proposed framework focuses on accurate fish segmentation from underwater images while learning robust feature representations that can be transferred to the subsequent tracking stage. The proposed segmentation network is trained using annotated images from the

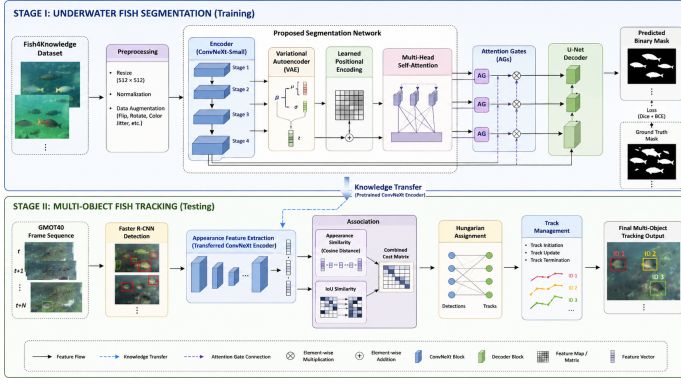


Fig. 1: Overall workflow of the proposed two-stage underwater fish segmentation and multi-object tracking framework.

Fish4Knowledge (F4K) dataset, where each RGB image is paired with a corresponding binary segmentation mask. Prior to training, all images and masks are resized to a fixed resolution of 512×512 pixels and normalized to ensure numerical stability during optimization. To improve the model's robustness against underwater imaging variations, data augmentation techniques including random horizontal flipping, vertical flipping, rotation, and color jittering are employed during training.

After preprocessing, the input image is forwarded to a ConvNeXt-Small encoder [3], which serves as the backbone for hierarchical feature extraction. Unlike conventional convolutional backbones, ConvNeXt adopts modernized convolutional design principles, including depthwise convolutions, inverted bottleneck layers, Layer Normalization, and GELU activation, enabling the extraction of rich semantic representations while maintaining computational efficiency. The encoder progressively transforms the input image into multi-scale feature maps, capturing both low-level spatial information and high-level semantic characteristics of underwater fish.

The extracted latent features are subsequently processed by a Variational Autoencoder (VAE) [4], which learns a probabilistic latent representation of the underwater scene. Instead of directly utilizing deterministic feature maps, the VAE estimates the latent mean (μ) and variance (σ), allowing latent variables to be sampled through the reparameterization strategy. This probabilistic representation improves the network's robustness to underwater appearance variations, illumination changes, and background noise while encouraging better feature generalization.

To preserve spatial information before contextual modeling, learned positional encoding is incorporated into the latent feature representation. The encoded features are then passed through a multi-head self-attention [5] module, where long-range dependencies among spatial locations are modeled by computing relationships between all feature positions. Unlike conventional convolutional operations that primarily capture local neighborhood information, the self-attention mechanism enables the network to exploit global contextual relationships,

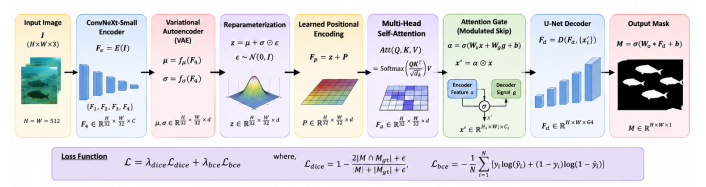


Fig. 2: Mathematical workflow of the proposed Stage I segmentation network illustrating feature extraction using ConvNeXt-Small, latent representation learning through the Variational Autoencoder, learned positional encoding, multi-head self-attention, attention-gated feature fusion, decoder reconstruction, and optimization using the combined Dice-BCE loss function.

thereby improving the segmentation of fish with varying shapes, orientations, and partially occluded boundaries.

Instead of directly propagating encoder features to the decoder through conventional skip connections, the proposed architecture employs attention gates to selectively filter encoder feature maps before feature fusion. Each attention gate suppresses irrelevant background activations while emphasizing salient fish regions, allowing only informative spatial features to be transferred to the decoder. This selective feature refinement improves boundary localization and reduces false-positive predictions, particularly in cluttered underwater environments.

Finally, the refined multi-scale features are progressively reconstructed by a U-Net decoder through successive upsampling and feature fusion operations to generate high-resolution binary segmentation masks. The entire segmentation network is optimized using a hybrid Dice Binary Cross-Entropy (Dice-BCE) loss function, which simultaneously improves pixel-wise classification accuracy and overlap between the predicted segmentation mask and the corresponding ground-truth annotation. The trained ConvNeXt-Small encoder obtained from this stage is subsequently transferred to the tracking framework as a discriminative appearance feature extractor, establishing the knowledge transfer strategy proposed in this work.

C. Knowledge Transfer Strategy

To establish a connection between the segmentation and tracking stages, the proposed framework employs a knowledge transfer strategy by reusing the pretrained ConvNeXt-Small encoder learned during Stage I. The encoder is trained on the Fish4Knowledge dataset to extract robust hierarchical feature representations that effectively distinguish fish from complex underwater backgrounds. Instead of training an independent appearance model, the learned encoder weights are directly transferred to the tracking framework, enabling the extraction of discriminative appearance embeddings for each detected fish.

Let D_i denote the i^{th} fish detected by the Faster R-CNN detector. The pretrained ConvNeXt-Small encoder, denoted by $E_p(\cdot)$, transforms each detected fish into a fixed-dimensional appearance embedding,

$$F_i = E_p(D_i), \quad (1)$$

where $F_i \in \mathbb{R}^d$ represents the appearance feature vector and d is the embedding dimension.

The visual similarity between two detected fish is computed using cosine similarity,

$$S_{\text{app}}(i, j) = \frac{F_i^T F_j}{\|F_i\|_2 \|F_j\|_2}, \quad (2)$$

where F_i and F_j denote the appearance embeddings of detections i and j , respectively.

To jointly exploit appearance and spatial information, the appearance similarity is combined with the Intersection over Union (IoU) of the corresponding bounding boxes to construct the association cost matrix,

$$C(i, j) = \lambda (1 - S_{\text{app}}(i, j)) + (1 - \lambda) (1 - \text{IoU}(i, j)), \quad (3)$$

where $\lambda \in [0, 1]$ is a weighting parameter that balances the contribution of appearance similarity and spatial overlap. The resulting cost matrix is subsequently utilized in the tracking stage for object association and identity preservation across consecutive image frames.

D. Stage II: Multi-Object Fish Tracking

The second stage of the proposed framework performs multi-object fish tracking on the GMOT40 dataset by integrating object detection, appearance feature extraction, and data association. Unlike conventional tracking approaches that train an independent appearance model, the proposed framework utilizes the pretrained ConvNeXt-Small encoder transferred from the segmentation stage to generate discriminative feature representations for each detected fish. This strategy enables efficient feature reuse while improving identity consistency across consecutive image frames.

Given an input frame sequence, fish instances are first detected using the Faster R-CNN detector. For the t^{th} frame, the detector predicts a set of bounding boxes,

$$\mathcal{D}_t = \{B_1, B_2, \dots, B_N\}, \quad (4)$$

where B_i denotes the bounding box corresponding to the i^{th} detected fish.

Each detected fish is subsequently forwarded to the pretrained ConvNeXt-Small encoder to obtain a discriminative appearance embedding as described in Section III-C. The appearance similarity and spatial overlap between detections and existing tracks are jointly considered to construct the association cost matrix. The spatial similarity is computed using the Intersection over Union (IoU),

$$\text{IoU}(i, j) = \frac{|B_i \cap B_j|}{|B_i \cup B_j|}, \quad (5)$$

where B_i and B_j denote the bounding boxes of the current detection and the existing track, respectively.

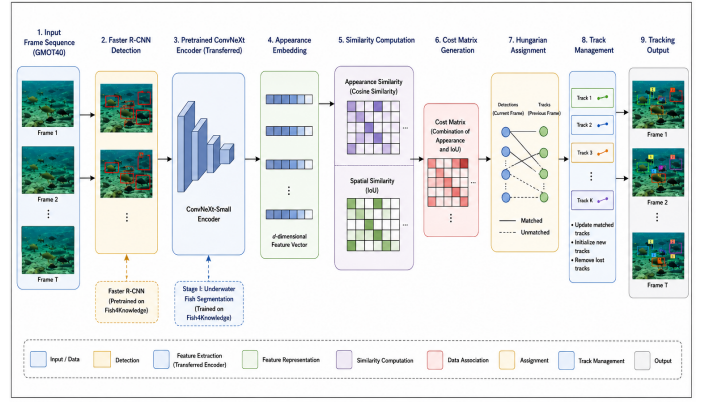


Fig. 3: Architecture of the proposed Stage II multi-object fish tracking framework.

The optimal correspondence between detections and existing tracks is obtained using the Hungarian assignment algorithm by minimizing the overall association cost,

$$\pi^* = \arg \min_{\pi} \sum_{i=1}^N C(i, \pi(i)), \quad (6)$$

where $C(i, \pi(i))$ represents the association cost defined in Eq. (3), and π denotes the optimal assignment between detections and active tracks.

Following data association, matched tracks are updated using the newly assigned detections, while unmatched detections initialize new tracks and inactive tracks are terminated after exceeding a predefined inactivity threshold. This track management strategy enables the framework to maintain consistent fish identities despite appearance variations, partial occlusions, and complex underwater environments. The final output of the tracking stage consists of bounding boxes with unique track identities for each fish throughout the image sequence.

IV. EXPERIMENTAL SETUP

A. Experimental Environment

All experiments were conducted on the *Lightning AI* cloud platform using an NVIDIA Tesla T4 GPU with 16 GB of GPU memory. The proposed framework was implemented in Python using the PyTorch deep learning framework and executed within a Jupyter Notebook environment. CUDA acceleration was employed to enable efficient model training and inference. Image preprocessing and data augmentation were performed using Albumentations and OpenCV, while Torchvision, NumPy, Pandas, Matplotlib, and Scikit-learn were utilized for feature processing, visualization, and performance evaluation.

B. Datasets

The proposed two-stage framework was evaluated using two publicly available underwater datasets. The Fish4Knowledge (F4K) dataset [11] was employed for the underwater fish segmentation task, while the GMOT40 dataset [12] was utilized to evaluate the multi-object fish tracking framework.

TABLE I: Hyperparameter settings of the proposed framework.

Stage	Parameter	Value
Segmentation	Platform	Lightning AI
	GPU	NVIDIA Tesla T4 (16 GB)
	Input Size	512×512
	Backbone	ConvNeXt-Small
	Optimizer	AdamW
	Epochs	50
Tracking	Detector	Faster R-CNN (ResNet-50 FPN)
	Optimizer	AdamW
	Epochs	10
	Confidence Threshold	0.50
	IoU Weight	0.70
	Appearance Weight	0.30
	Association	Hungarian Algorithm

The Fish4Knowledge dataset contains underwater fish images captured under diverse marine environments with significant variations in illumination, water turbidity, fish species, object scale, pose, and background complexity. Each RGB image is accompanied by a corresponding binary segmentation mask, which serves as the ground-truth annotation for supervised segmentation training.

The GMOT40 dataset consists of densely populated image sequences containing multiple objects with frame-wise bounding box annotations. In this work, only the fish category of the GMOT40 dataset was considered for evaluating the proposed tracking framework. The dataset presents several challenges, including occlusion, abrupt object motion, appearance similarity among different fish, and complex underwater backgrounds, making it suitable for assessing multi-object tracking performance.

C. Training Configuration

The proposed framework was trained in two sequential stages. During Stage I, the segmentation network was trained on the Fish4Knowledge dataset using the AdamW optimizer [16] with a hybrid Focal-Dice [13], [14] and Kullback-Leibler (KL) divergence loss. After training, the pretrained ConvNeXt-Small encoder was transferred to the tracking framework. During Stage II, Faster R-CNN with a ResNet-50-FPN backbone [8], [9] was employed for fish detection on the GMOT40 dataset. The extracted appearance embeddings, together with IoU-based spatial similarity, were used for data association through the Hungarian assignment algorithm. The principal training hyperparameters used in both stages are summarized in Table I.

D. Evaluation Metrics

The segmentation performance was evaluated using Accuracy, Precision, Recall, F1-score, Dice Similarity Coefficient (DSC), and Intersection over Union (IoU). These metrics quantify the pixel-wise agreement between the predicted segmentation masks and the corresponding ground-truth annotations.

The tracking performance was evaluated using standard multi-object tracking metrics, including Multiple Object

TABLE II: Performance comparison on the Fish4Knowledge dataset.

Method	Acc.	Prec.	Rec.	F1	IoU
Baseline (U-Net)	83.7	62.1	64.7	63.4	65.1
Self Attention	89.3	83.6	86.3	85	71.2
Previous Model (VAE + SA)	92	89.3	88.3	88.8	78.4
Proposed Framework	95.46	93.12	97.93	95.46	91.32

Tracking Accuracy (MOTA), Multiple Object Tracking Precision (MOTP), Identity F1-score (IDF1), Identity Switches (IDSW), Mostly Tracked (MT), Mostly Lost (ML), and Fragmentation (Frag). These metrics comprehensively assess detection accuracy, identity preservation, and overall tracking robustness in challenging underwater environments.

V. RESULTS AND DISCUSSION

A. Segmentation Results

The segmentation performance of the proposed framework was evaluated on the Fish4Knowledge dataset using standard pixel-level evaluation metrics, including Accuracy, Precision, Recall, F1-score, Dice Similarity Coefficient (DSC), and Intersection over Union (IoU). The proposed architecture combines a ConvNeXt-Small encoder, Variational Autoencoder, learned positional encoding, multi-head self-attention, and attention-gated skip connections to achieve robust segmentation under challenging underwater conditions. Both quantitative and qualitative evaluations were performed to demonstrate the effectiveness of the proposed approach.

1) *Qualitative Results*: Figure 4 presents representative segmentation results produced by the proposed framework on the Fish4Knowledge dataset. Each row corresponds to a different underwater image and includes the input image, ground-truth mask, predicted segmentation mask, and the overlay of the predicted mask on the original image. The visual results demonstrate that the proposed model accurately delineates fish boundaries while effectively suppressing complex underwater backgrounds. The integration of the ConvNeXt-Small encoder, Variational Autoencoder (VAE), multi-head self-attention, and attention-gated skip connections enables robust segmentation under varying illumination conditions, object scales, and background clutter.

B. Multi-Object Tracking Results

The tracking performance of the proposed framework was evaluated on the GMOT40 dataset using standard multi-object tracking (MOT) evaluation metrics. By transferring the pretrained ConvNeXt-Small encoder from the segmentation stage, the framework learns discriminative appearance representations that improve object association and identity preservation across consecutive image frames. The combination of appearance similarity and IoU-based spatial matching enables robust tracking under challenging underwater conditions, including occlusions, appearance variations, and complex backgrounds.

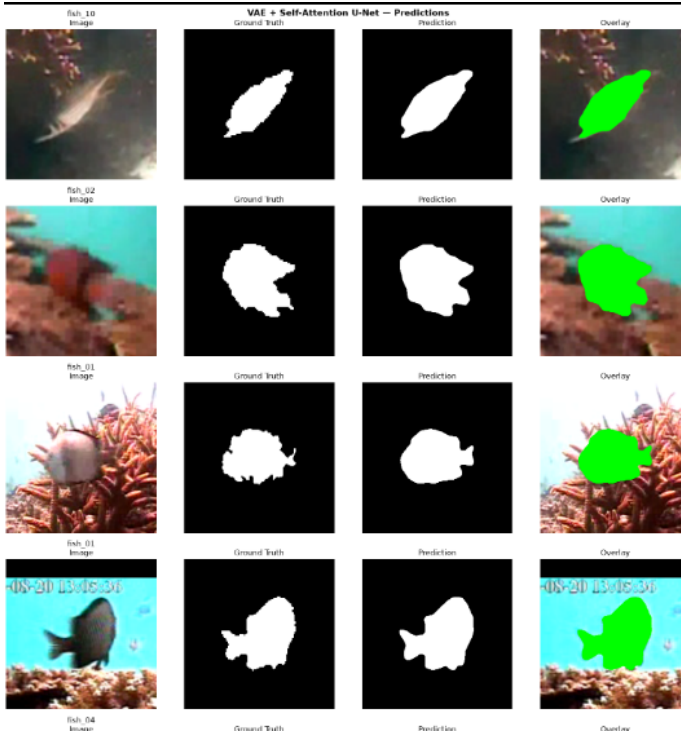


Fig. 4: Qualitative segmentation results of the proposed framework on the Fish4Knowledge dataset. From left to right: input image, ground-truth mask, predicted segmentation mask, and overlay of the predicted mask on the original image. The proposed method accurately localizes fish regions and preserves object boundaries under diverse underwater conditions.

TABLE III: Comparison of multi-object tracking performance on the GMOT40 dataset. Higher values of MT, MOTA, and MOTP indicate better performance, whereas lower values of ML, IDSW, and FM are preferred.

Tracker	GT	MT	PT	ML	IDSW	FM	MOTA	MOTP
Viou	290	180	39	71	64	48	76.0	74.0
IoU	290	183	45	62	52	42	76.8	73.0
MDP	290	164	104	22	212	110	77.0	89.6
DeepSORT	290	200	51	39	173	31	81.2	91.2
Mukherjee	290	110	123	57	50	34	79.3	78.0
Proposed	171	141	25	5	20	41	88.8	84.4

VI. CONCLUSION AND FUTURE WORK

This paper presented a two-stage underwater fish analysis framework that integrates fish segmentation and multi-object tracking through a knowledge transfer strategy. In the first stage, a novel segmentation network was developed by combining a ConvNeXt-Small encoder, Variational Autoencoder (VAE), learned positional encoding, multi-head self-attention, attention-gated skip connections, and a U-Net decoder to generate accurate binary fish segmentation masks. The pretrained ConvNeXt-Small encoder was subsequently transferred to the tracking stage, where it served as a discriminative appearance

feature extractor for robust multi-object fish tracking on the GMOT40 dataset. By combining appearance similarity with IoU-based spatial matching and Hungarian assignment, the proposed framework effectively maintained object identities across consecutive image frames.

Experimental results on the Fish4Knowledge and GMOT40 datasets demonstrate the effectiveness of the proposed framework. The segmentation model achieved accurate fish localization under challenging underwater conditions, while the tracking framework improved identity preservation and achieved competitive multi-object tracking performance compared with existing approaches. The proposed knowledge transfer strategy also reduced redundant feature learning by enabling a shared feature representation between segmentation and tracking.

Future work will focus on extending the proposed framework to support real-time underwater monitoring by incorporating lightweight network architectures and efficient tracking algorithms. In addition, transformer-based temporal feature learning and adaptive data association strategies will be investigated to further improve tracking accuracy under severe occlusions and dense fish populations. The framework will also be evaluated on larger and more diverse underwater datasets containing multiple fish species and varying environmental conditions to enhance its robustness and generalization capability for practical marine monitoring applications.

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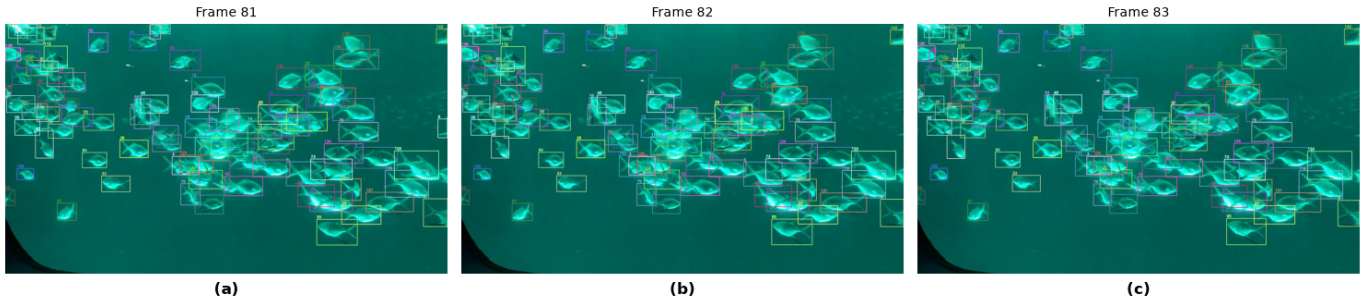


Fig. 5: Qualitative tracking results on consecutive frames of the GMOT40 dataset. Each fish is assigned a unique identity that is consistently maintained across successive frames, demonstrating robust multi-object tracking in complex underwater environments.

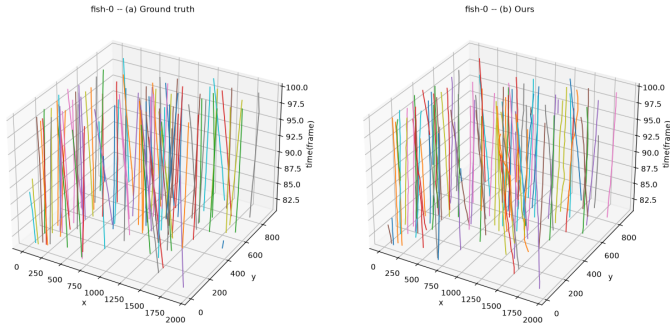


Fig. 6: Comparison of ground-truth and predicted fish trajectories on the GMOT40 dataset. The proposed tracking framework accurately preserves the temporal trajectories of multiple fish, demonstrating effective identity preservation and robust data association across consecutive image frames.

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